Exercise-01: User Authentication Script

File name: **“hello.sh”**

Source code:

```
#! /bin/bash

data="userdata.txt"
read -p "Enter username: " name
echo -n "Enter Password: "
read -s pass
echo ""

join="$name $pass"

f=0
while read line;
do
if [ "$join" == "$line" ]; then
f=1
echo "welcome to your account"
break
fi
done < "$data"

if [ $f -eq 0 ]; then
echo "Invalid password or username"
fi
```

Output:

1. *Prompt the user to enter a username:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./hello.sh
Enter username: alice
```

2. *Prompt the user to enter a password:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./hello.sh
Enter username: alice
Enter Password:
```

3. <u>*userdata.txt file contents*</u>

```
 GNU nano 7.2                         userdata.txt
alice 12345
bob qwerty
charlie pass123
adil 7894
```

4. <u>*If the entered username and password pair matches an entry in the file → display a welcome message.*</u>

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./hello.sh
Enter username: alice
Enter Password:
welcome to your account
```

5. <u>*If it does not match → display an error message.*</u>

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./hello.sh
Enter username: Alice
Enter Password:
Invalid password or username
```

Exercise-02: Fun Todo List Manager

File name: **"todo.sh"**

Source Code:

```
#! /bin/bash

mark="[]"
complete="[x]"
tasks=""
file="todo.txt"
count=1
> update.txt

while read line;
do
if [ "$1" == "list" ]; then
echo "$line"
fi
task="$line"
```

```
if [[ "${line,,}" == *"${1,,}"* ]]; then
echo "$line"
fi

if [[ "$1" == "done" && $2 -eq $count ]]; then
task=""
echo "Nice job 🚀 "
for i in $line;
do

if [ "$i" == "$mark" ]; then
task="$task $complete"
else
task="$task $i"
fi

done
fi
echo "$task" >> update.txt
count=$((count+1))
done <"$file"

if [ "$1" == "add" ]; then
task="$((count)). $mark $2"
echo "task is created"
echo "$task" >> todo.txt
echo "$task" >> update.txt
fi


if [ "$1" == "clear" ]; then
> update.txt
echo "Todo list cleared"
fi

cp update.txt todo.txt
```

Output:

1. *Add Tasks to todo list:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh list
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh add "Read Bash note
s"
task is created
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh add "Finish OS Lab"
task is created
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh add "Submit Assignm
ent"
task is created
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh list
1. [] Read Bash notes
2. [] Finish OS Lab
3. [] Submit Assignment
```

2. *List tasks:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh list
1. [] Read Bash notes
2. [] Finish OS Lab
3. [] Submit Assignment
adil@ASUS-Vivobook-Intel-corei7:~$
```

3. *Mark as done:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh done 2
Nice job 🚀
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh list
1. [] Read Bash notes
2. [x] Finish OS Lab
3. [] Submit Assignment
```

4. *Search task by keyword:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh "lab"
2. [x] Finish OS Lab
adil@ASUS-Vivobook-Intel-corei7:~$
```

5. *Clear all the tasks:*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh clear
Todo list cleared
adil@ASUS-Vivobook-Intel-corei7:~$ ./todo.sh list
adil@ASUS-Vivobook-Intel-corei7:~$
```